

Formal Languages and Finite Automata
Homework Assignment #4

1. (30%) Construct an npda to accept the language: $L = \{a^{3n}b^{2n} : n \geq 0\}$. Do not use the CFG-to-NPDA conversion procedure.
2. (40%) Given the following npda $M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$, where $Q = \{q_0, q_1, q_2\}$, $\Sigma = \{a, b\}$, $\Gamma = \{z, A\}$, $F = \{q_2\}$ and transitions

$$\delta(q_0, a, z) = \{(q_0, Az)\}$$

$$\delta(q_0, a, A) = \{(q_0, AA)\}$$

$$\delta(q_0, b, A) = \{(q_1, \lambda)\}$$

$$\delta(q_1, b, A) = \{(q_1, \lambda)\}$$

$$\delta(q_1, \lambda, z) = \{(q_2, \lambda)\}$$

Convert the npda into an equivalent CFG using the procedure introduced in class, remove all useless variables, and use A_{ij} instead of $q_i A q_j$.

3. (30%) Construct a three-state npda for the language generated by this grammar:
 $S \rightarrow aAb \mid bBa$
 $A \rightarrow B \mid a$
 $B \rightarrow bB \mid b$